

E-Commerce Data Analytics Project Report

1 Business Problem

- **Uncleaned Data Bottlenecks:** Addressed raw data inconsistencies, missing values, and formatting errors to create a clean, reliable source of truth.
- **Logistics & Fulfillment Blind Spots:** Monitored order statuses and delivery durations to uncover why orders face cancellations or returns.
- **Revenue & Profitability Tracking:** Quantified exact net revenue and profit margins across different product categories and payment modes.

2 Tech Stack Summary

- **Python (Pandas, NumPy):** Used for data cleaning, preprocessing, and feature engineering.
- **Excel:** Initial data source handling and exploration.
- **MySQL & MySQL Workbench:** Database management, storing records, and exploratory data analysis.
- **Power BI:** Interactive dashboards, Power Query transformations, DAX measures, and TMDL modeling.

3 Workflow in Detail

- **Step 1 - Data Preparation (Excel & Python):** Started with raw Excel datasets, handled missing values, removed duplicate entries, and standardized text formatting using Pandas. Engineered core financial metrics (Sales, Net Account, Profit) and extracted temporal attributes (Order_Month, Order_Year, Weekday).
- **Step 2 - Database Integration (MySQL):** Loaded cleaned records into a MySQL database via SQLAlchemy to structure the data for 333 total orders and execute exploratory queries.
- **Step 3 - Business Intelligence & Modeling (Power BI):** Connected MySQL Workbench to Power BI, performed Power Query transformations, and implemented custom columns and DAX measures (such as Month Numbers for chronological sorting, Delivery Days, and Delivery Status) to evaluate logistics performance and sales trends.

4 Business Insights

- **Financial Performance:** Recorded 280 total orders, generating \$2.84M in Total Revenue, \$567.02K in Total Profit, a consistent 20% Profit Margin, and an AOV of 10.13K.
- **Category Profitability & Units Sold:** Electronics led profitability at \$0.23M and highest units sold (37.2%), followed by Fashion (\$0.14M), Furniture (\$0.10M), and Home (\$0.09M).

- **Order Status Breakdown:** Tracked order fulfillment distribution across Cancelled (27.5%), Returned (25.71%), Delivered (24.64%), and Pending (22.14%).
- **Delivery Status vs. Order Status Logistics:** Mapped delivery performance across delivery states (Late vs. On Time) against order statuses (Cancelled, Delivered, Pending, Returned), highlighting that delayed fulfillment heavily coincides with high cancellation and return volumes.
- **Monthly & Quarterly Trends:** Monthly profit peaked in February (\$0.12M) and May (\$0.13M), while quarterly distribution recorded Q1 at 129 orders (46.07%) and Q2 at 151 orders (53.93%).

5 Impact of the Analytics Project on the Business

- **Driven by Financial Visibility:** Established rigorous tracking of core financial performance (\$2.84M in revenue and \$567.02K in profit) with a stable 20% profit margin and an AOV of 10.13K, providing management with reliable financial metrics for forecasting and budgeting.
- **Optimized Category Strategy:** Highlighted high-performing drivers—specifically Electronics (leading with \$0.23M in profit and 353 units sold) and key items like Laptops and Mice—enabling targeted inventory planning and marketing resource allocation.
- **Risks Analysis:** Uncovered high cancellation (27.5%) and return rates tied directly to logistics and delivery delays, offering clear operational visibility to fix supply chain bottlenecks and improve on-time delivery execution.

6 Dashboard Overview

Here is the visual representation of the project analytics:

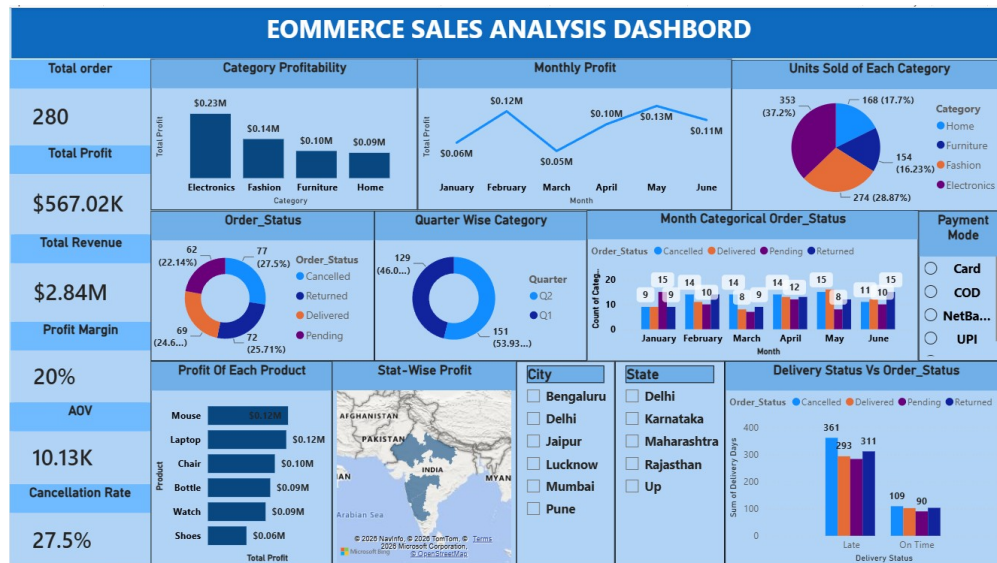


Figure 1: E-Commerce Sales Analysis Dashboard